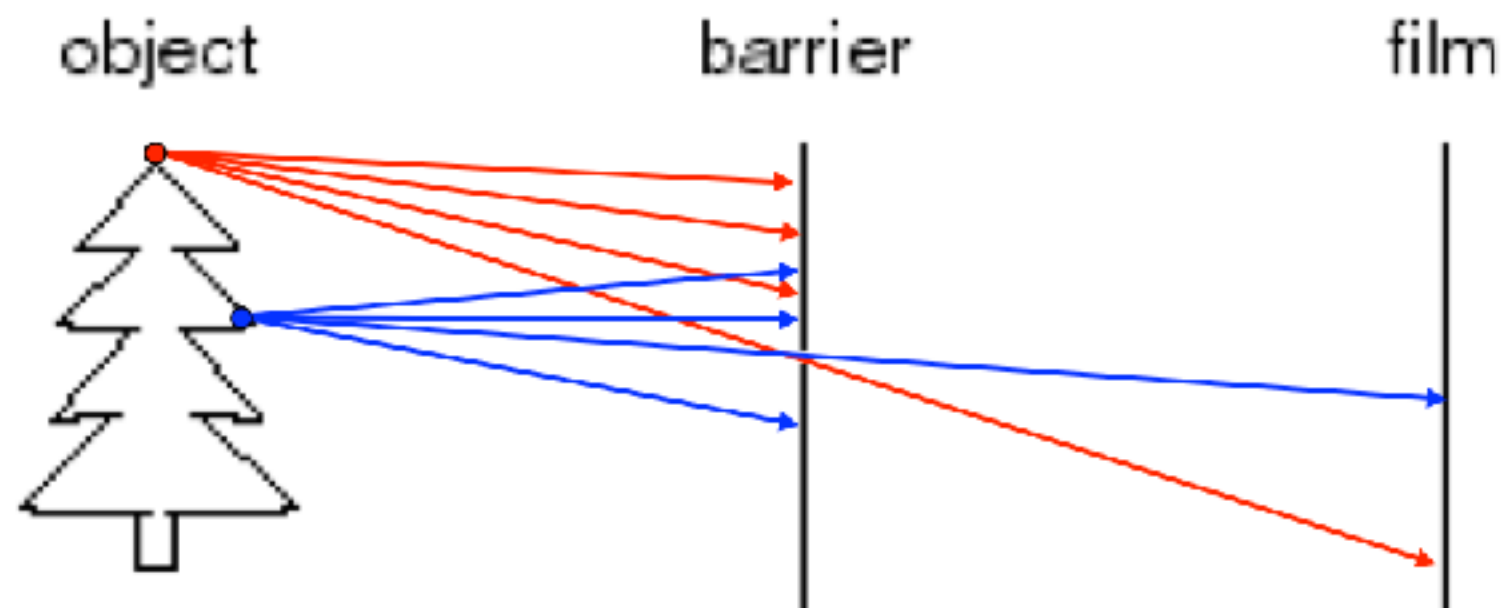


Last week: The pinhole camera

Pinhole camera

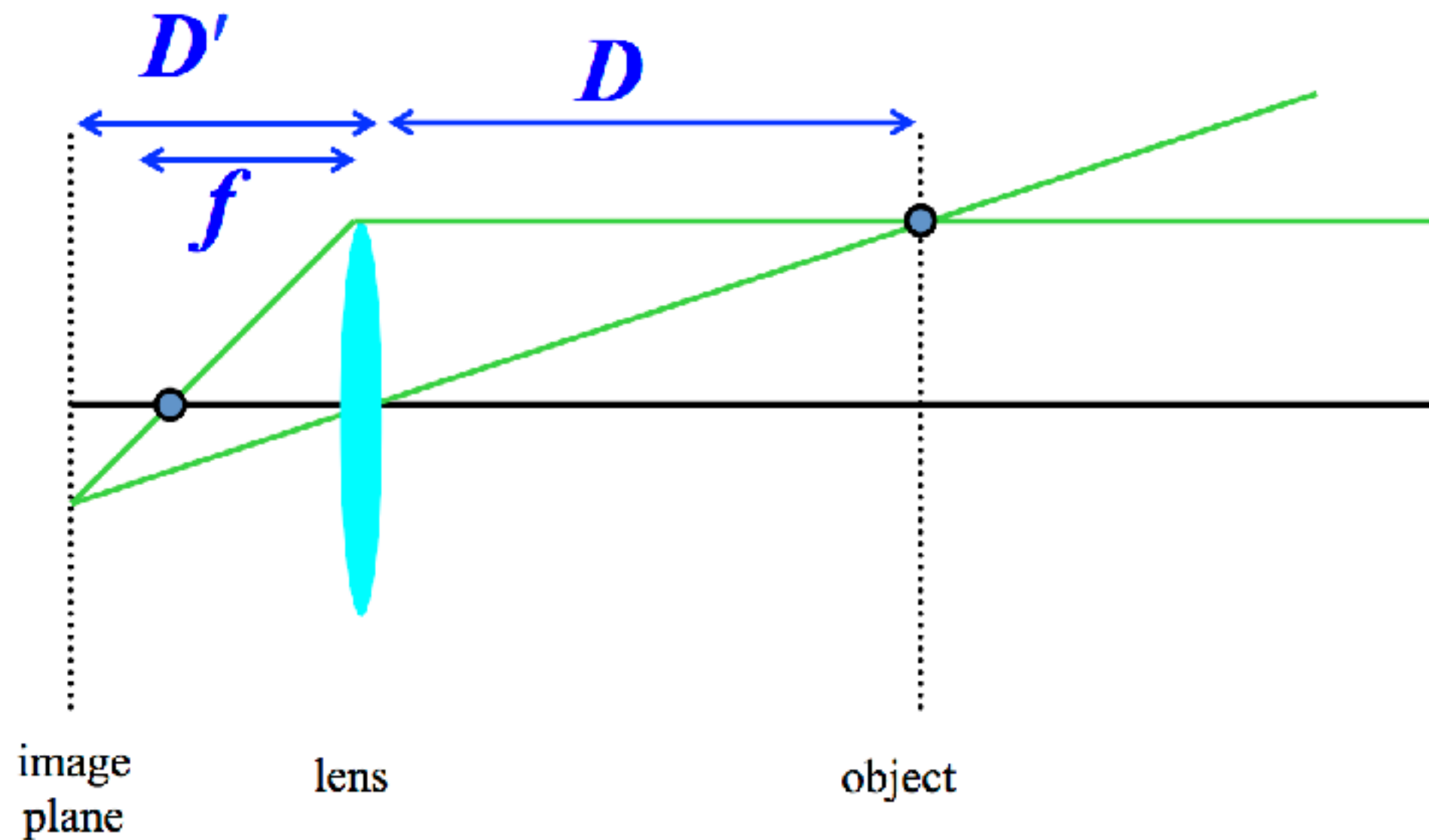


Add a barrier to block off most of the rays

Thin lens formula

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Any point satisfying the thin lens equation is in focus.



Projecting a point

Thus we map $(x, y, z) \rightarrow (x \frac{-d}{z}, y \frac{-d}{z})$.

Using homogeneous coordinates, we can write

$$\begin{bmatrix} x \\ y \\ -z/d \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1/d & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

This is equivalent to the normalized coordinate in the Cartesian image plane:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x \cdot \frac{-d}{z} \\ y \cdot \frac{-d}{z} \\ 1 \end{bmatrix} \approx \begin{bmatrix} x \\ y \\ -\frac{z}{d} \end{bmatrix}$$

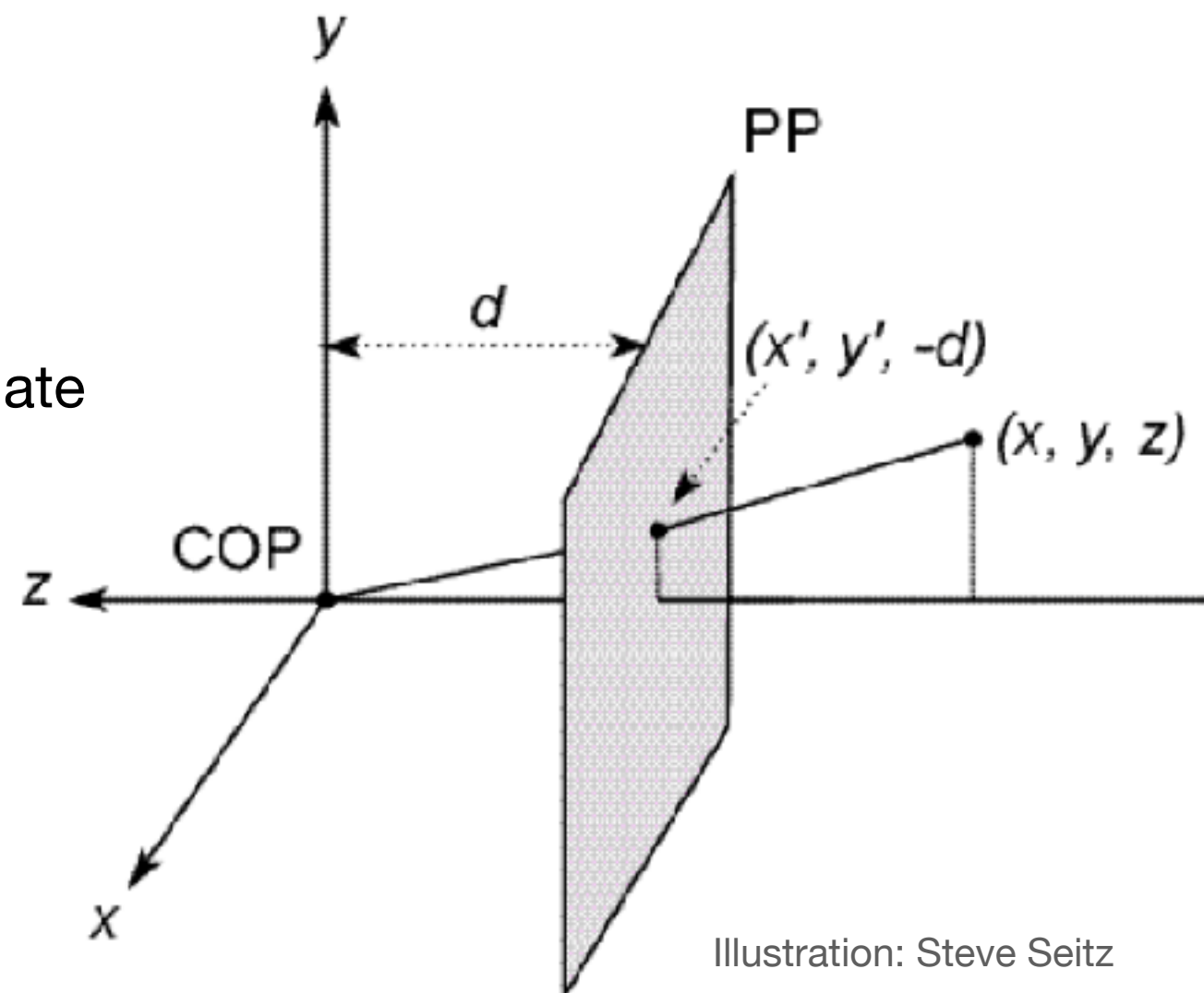


Illustration: Steve Seitz

Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \Rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

divide by the third coordinate

In practice: lots of coordinate transformations...

$$\begin{pmatrix} \text{2D} \\ \text{point} \\ (3 \times 1) \end{pmatrix} = \begin{pmatrix} \text{Camera to} \\ \text{pixel coord.} \\ \text{trans. matrix} \\ (3 \times 3) \end{pmatrix} \begin{pmatrix} \text{Perspective} \\ \text{projection matrix} \\ (3 \times 4) \end{pmatrix} \begin{pmatrix} \text{World to} \\ \text{camera coord.} \\ \text{trans. matrix} \\ (4 \times 4) \end{pmatrix} \begin{pmatrix} \text{3D} \\ \text{point} \\ (4 \times 1) \end{pmatrix}$$

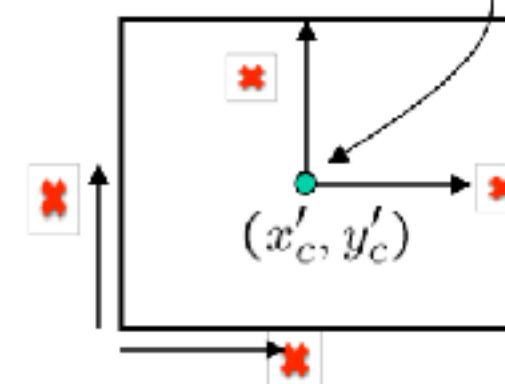
Camera parameters

A camera is described by several parameters

- Translation $\mathbf{T}$ of the optical center from the origin of world coords
- Rotation $\mathbf{R}$ of the image plane
- focal length f , principle point (x'_c, y'_c) , pixel size (s_x, s_y)
- blue parameters are called “extrinsics,” red are “intrinsics”

Projection equation

$$\mathbf{X} = \begin{bmatrix} sx \\ sy \\ s \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{\Pi} \mathbf{X}$$



- The projection matrix models the cumulative effect of all parameters
- Useful to decompose into a series of operations

$$\mathbf{\Pi} = \underbrace{\begin{bmatrix} -fs_x & 0 & x'_c \\ 0 & -fs_y & y'_c \\ 0 & 0 & 1 \end{bmatrix}}_{\text{intrinsics}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{projection}} \underbrace{\begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{rotation}} \underbrace{\begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{translation}}$$

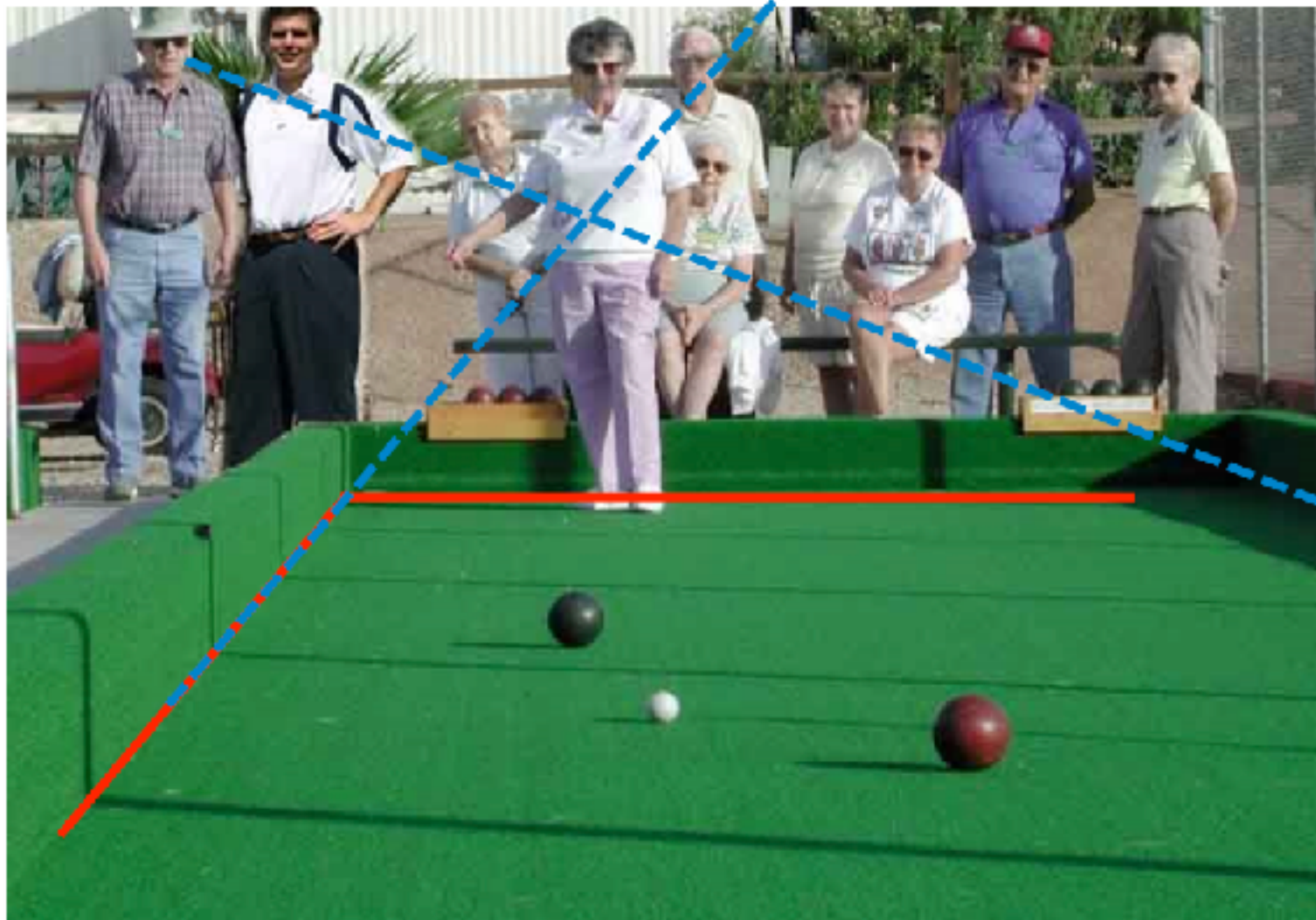
identity matrix

- The definitions of these parameters are **not** completely standardized
 - especially intrinsics—varies from one book to another

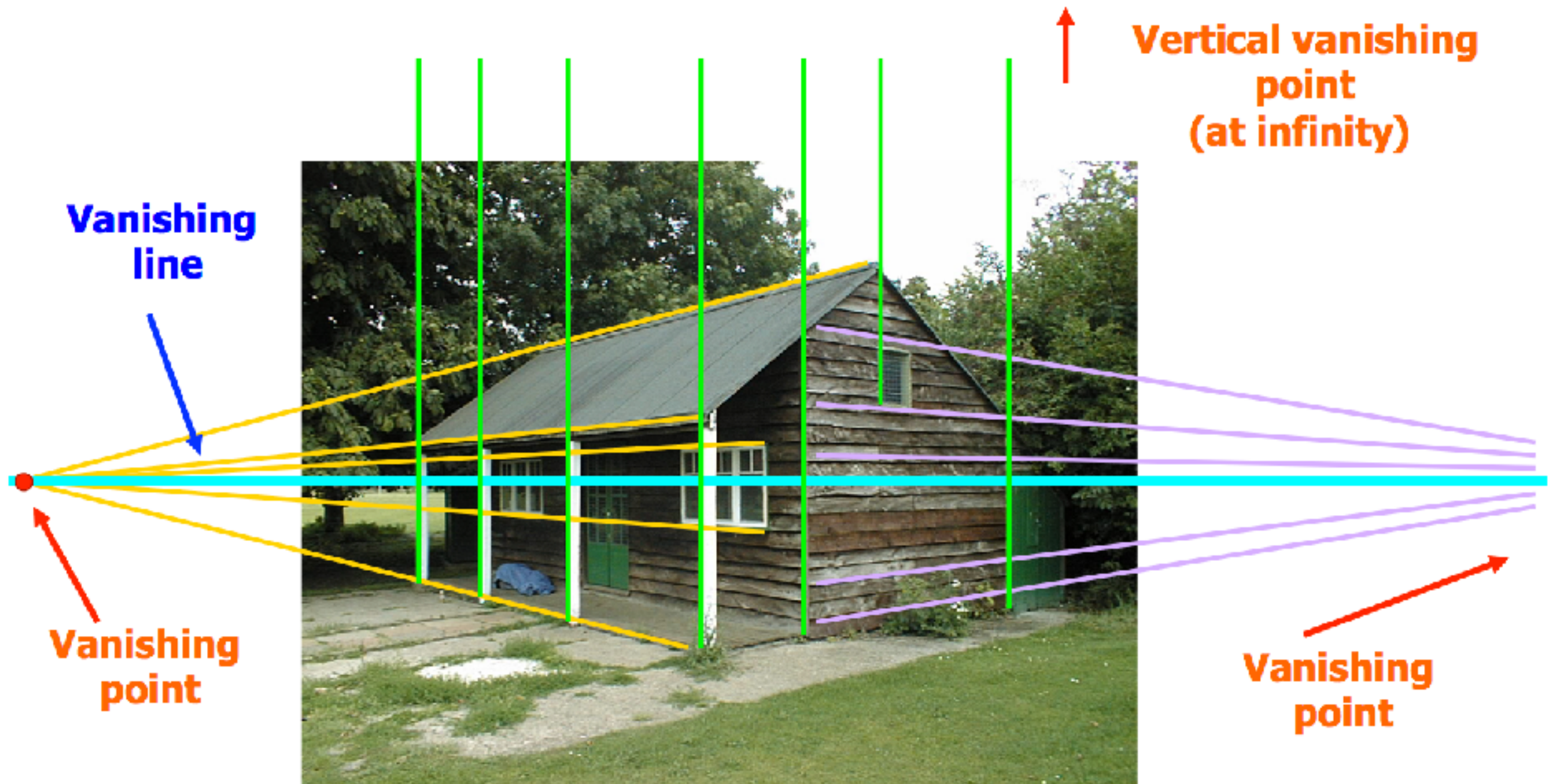
Projective Geometry

What is preserved?

- Straight lines are still straight



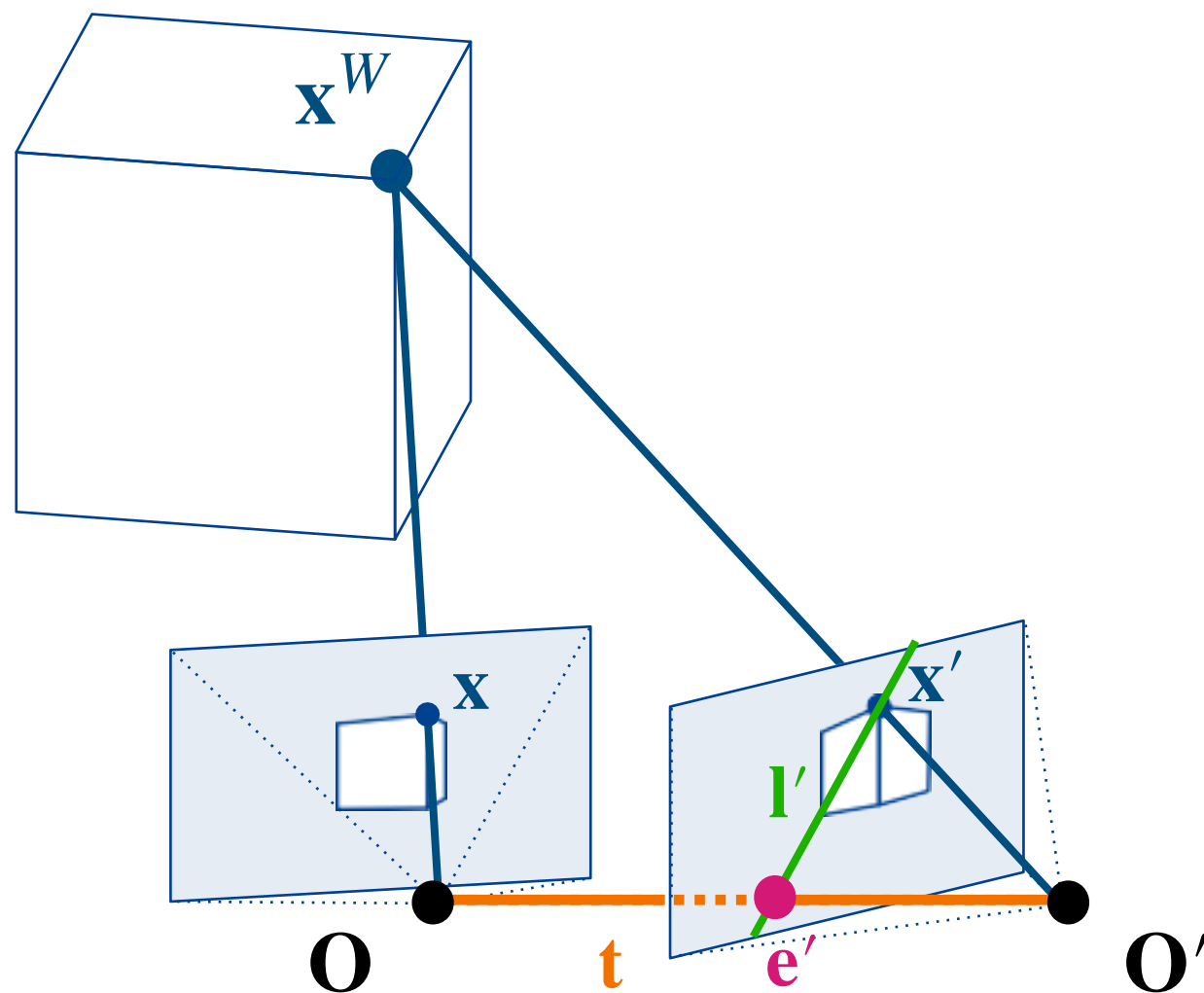
Vanishing points and lines



Last week: Epipolar Geometry

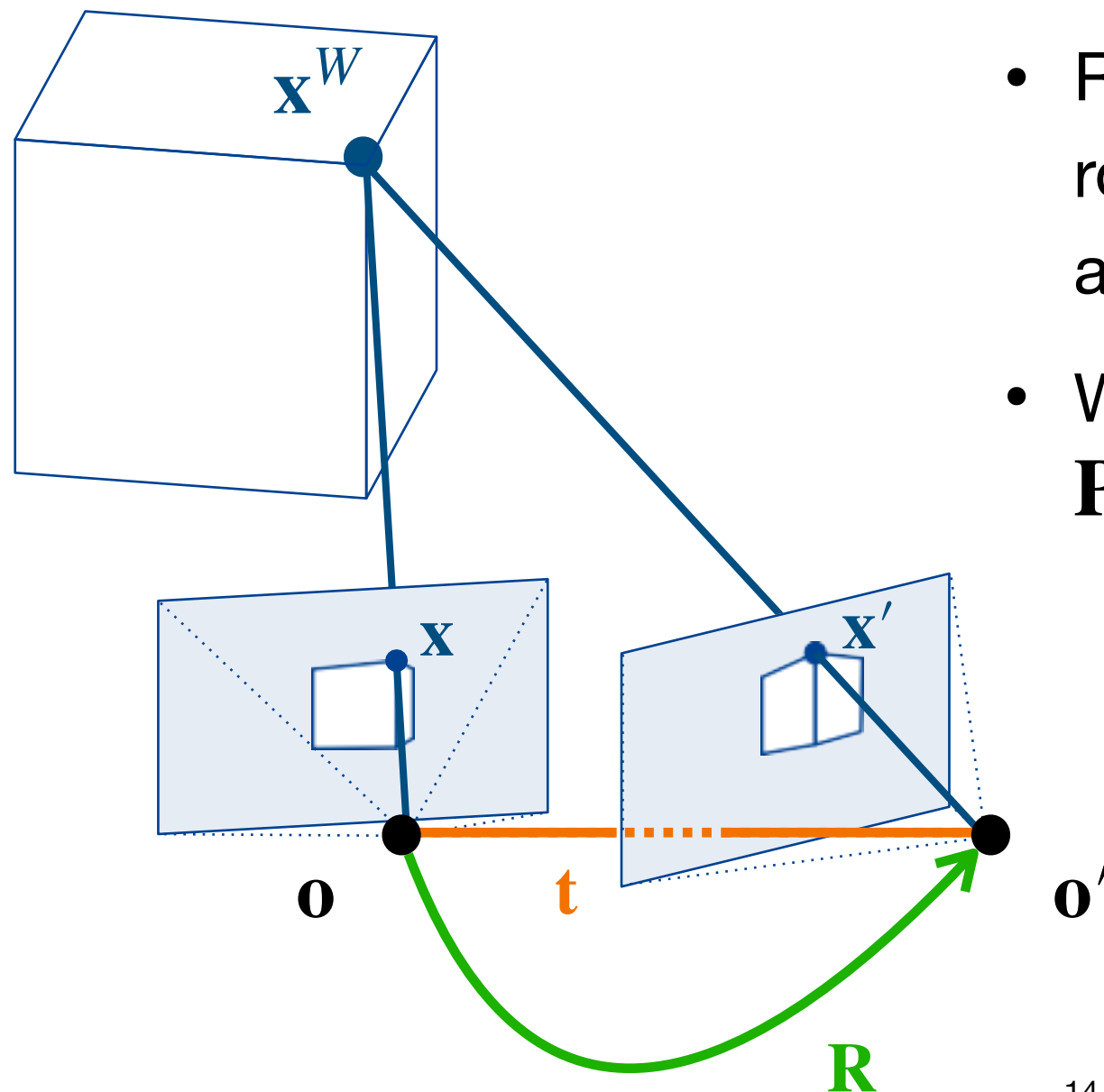
Epipolar geometry

- $\mathbf{o}, \mathbf{o}'$: projection centers
- $\mathbf{x}, \mathbf{x}'$: image points of $\mathbf{x}^W$
- $\mathbf{t}$: stereo basis
- $\mathbf{e}'$: right epipole
(Projection of $\mathbf{o}$)
- $\mathbf{l}'$: epipolar line
Connection of $\mathbf{e}'$ with $\mathbf{x}'$; contains all points observed as $\mathbf{x}$ in the left view



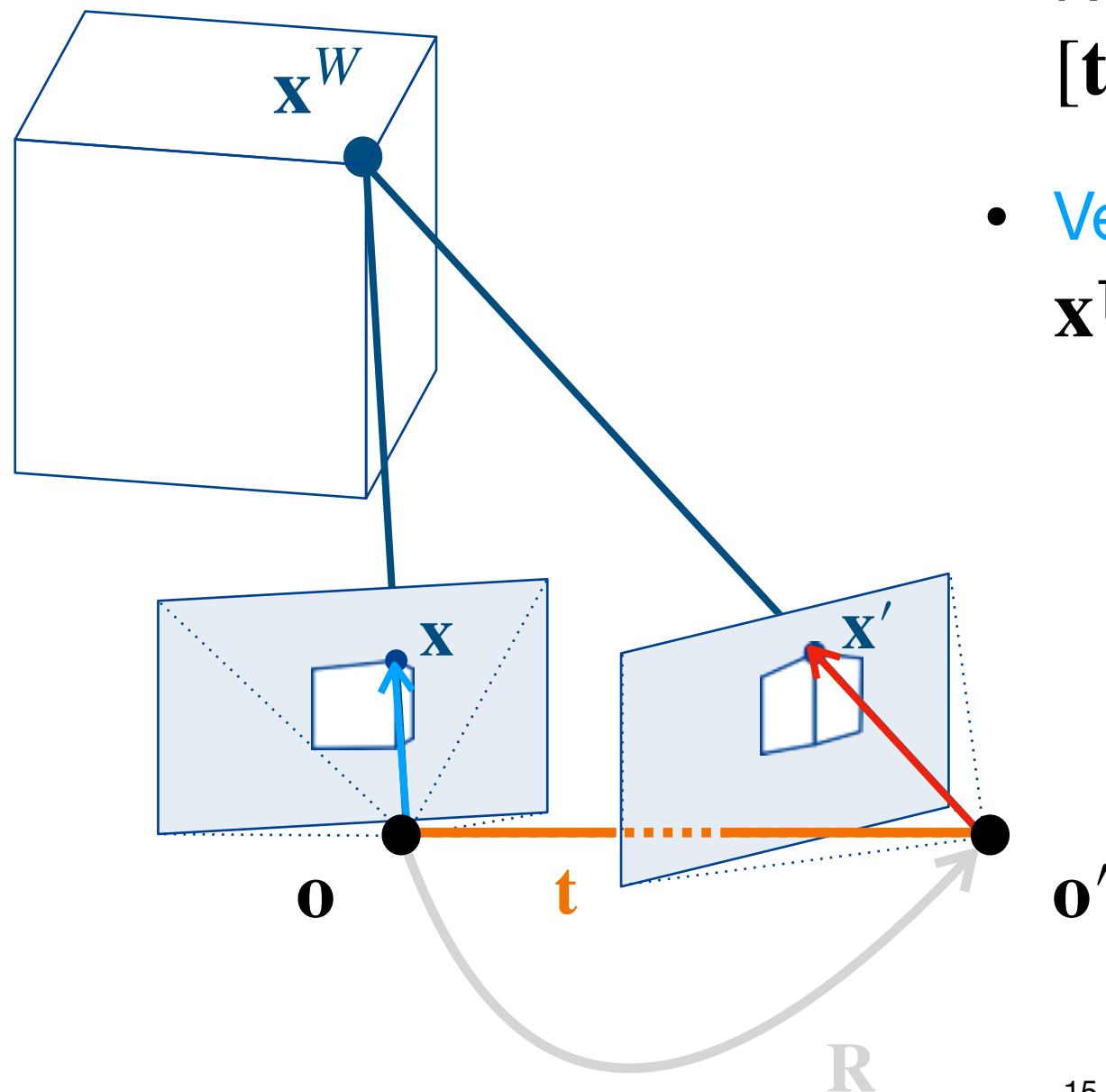
Simplified coordinate systems

- Canonical Kameras: $\mathbf{K} = \mathbf{K}' = \mathbf{I}$
- World coordinate system in the left camera system
- Right Kamera shifted by $\mathbf{t}$ and rotated by $\mathbf{R}$ - points $\mathbf{x}^W$ will thus appear as $\mathbf{R}^T(\mathbf{x}^W - \mathbf{t})$
- We obtain projection matrices $\mathbf{P} = [\mathbf{I} \quad \mathbf{0}]$ und $\mathbf{P}' = [\mathbf{R}^T \quad -\mathbf{R}^T\mathbf{t}]$



The essential matrix E

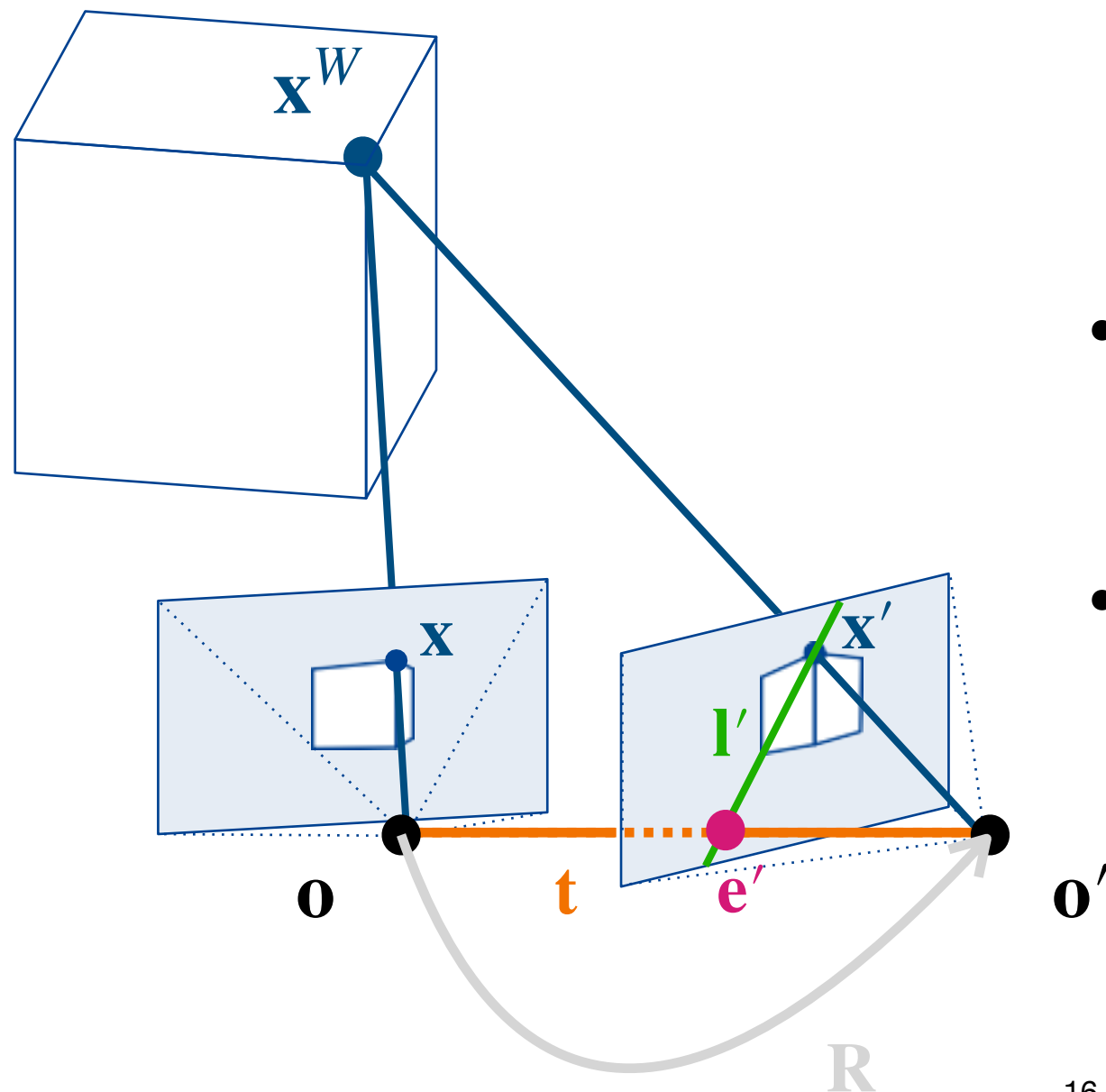
We span the epipolar plane in the coordinate system of the left camera.



- The **red vector** is $\mathbf{R} \mathbf{x}'$
- Normal of the epipolar plane:
 $[\mathbf{t}]_{\times} (\mathbf{R} \mathbf{x}')$
- **Vector $\mathbf{x}$** lies in the plane, hence
 $\mathbf{x}^T [\mathbf{t}]_{\times} \mathbf{R} \mathbf{x}' = 0$

Essential matrix
 $\mathbf{E} = [\mathbf{t}_{\times}] \mathbf{R}$
with epipolar condition
 $\mathbf{x}^T \mathbf{E} \mathbf{x}' = 0$

Properties of $\mathbf{E}$



- $\mathbf{x}'$ lies on the **epipolar line $\mathbf{l}'$** , hence $\mathbf{l}'^T \mathbf{x}' = 0$
- With $\mathbf{x}^T \mathbf{E} \mathbf{x}' = 0$ we see that $\mathbf{l}' = (\mathbf{x}^T \mathbf{E})^T = \mathbf{E}^T \mathbf{x}$ and $\mathbf{l} = \mathbf{E} \mathbf{x}'$
- In particular, $\mathbf{E} \mathbf{e}' = \mathbf{0}$ and $\mathbf{E}^T \mathbf{e} = \mathbf{0}$
- Due to $\mathbf{t}^T \mathbf{E} = \mathbf{t}^T [\mathbf{t}_\times] \mathbf{R} = 0$ we have $\mathbf{t}$ as the null space!

The fundamental matrix $\mathbf{F}$

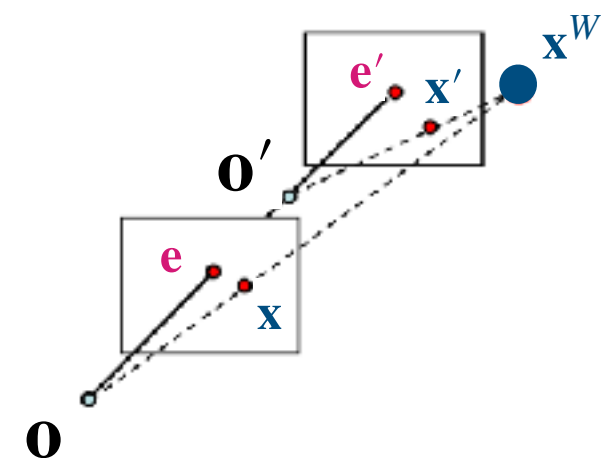
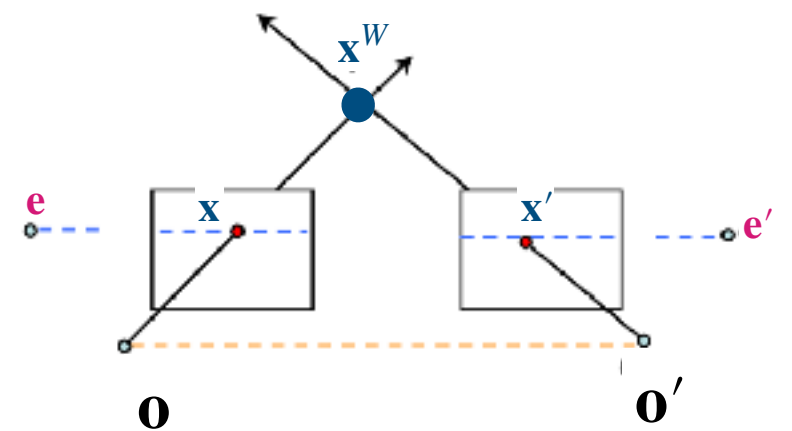
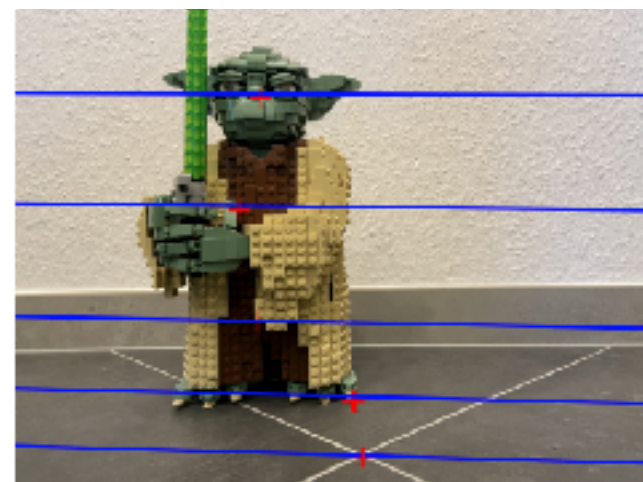
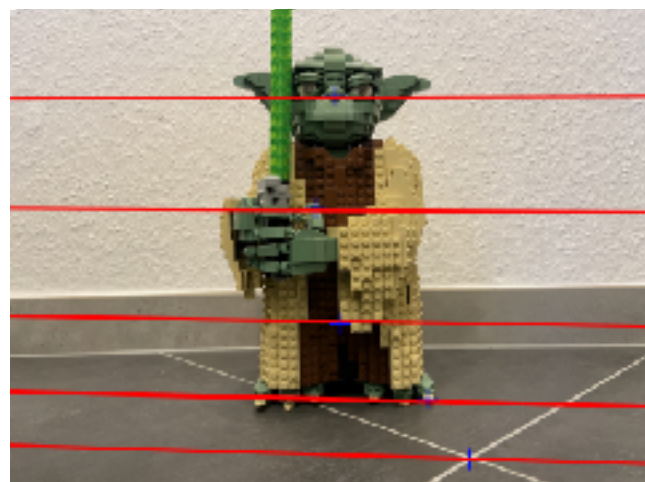
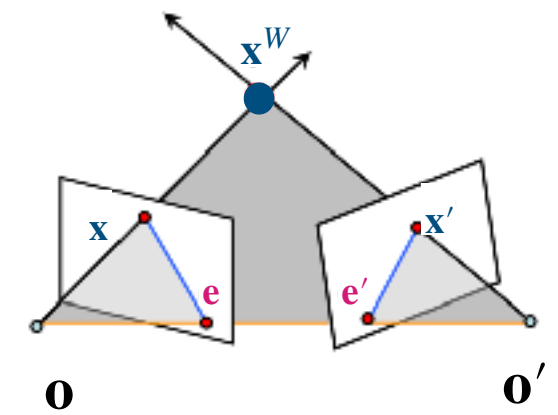
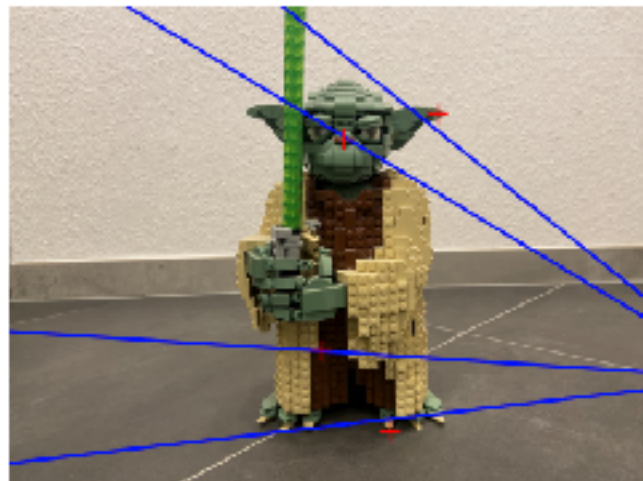
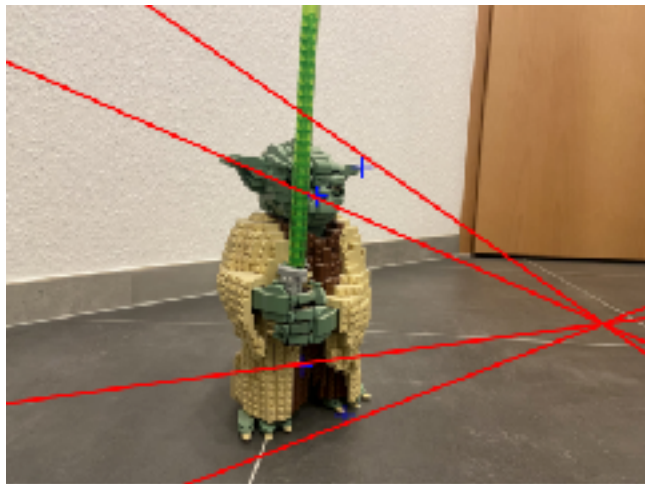
- Non-canonical pixel coordinates $\mathbf{p}, \mathbf{p}'$ can be converted using the intrinsic parameters to get $\mathbf{x} = \mathbf{K}^{-1}\mathbf{p}$ and $\mathbf{x}' = \mathbf{K}'^{-1}\mathbf{p}'$
- Then we have $\mathbf{x}^\top \mathbf{E} \mathbf{x}' = (\mathbf{K}^{-1}\mathbf{p})^\top \mathbf{E} \mathbf{K}'^{-1}\mathbf{p}' = \mathbf{p}^\top \mathbf{K}^{-\top} \mathbf{E} \mathbf{K}'^{-1} \mathbf{p}'$
- Thus there is a matrix $\mathbf{F}$ with $\mathbf{p}^\top \mathbf{F} \mathbf{p}' = 0$

Fundamental Matrix

$$\mathbf{F} = \mathbf{K}^{-\top} [\mathbf{t}_\times] \mathbf{R} \mathbf{K}'^{-1}$$

$$\mathbf{p}^\top \mathbf{F} \mathbf{p}' = 0$$

Different configurations



How to compute the
fundamental matrix F

Compute F from corresponding points

Wir haben $[u \ v \ 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = 0$ und stellen um:

$$[uu' \ uv' \ u \ vu' \ vv' \ v \ u' \ v' \ 1] \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix} = 0$$

The 8-point algorithm

- Construct the $N \times 9$ matrix $\mathbf{A}$ from corresponding points in the two images
- Calculate the singular value decomposition (SVD)
 $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$
- Choose the column in $\mathbf{V}$ corresponding to the smallest Eigenvalue in $\mathbf{S}$

The normalized 8-point algorithm

- Normalize the point correspondences
- Construct the $M \times 9$ design matrix $\mathbf{A}$
- Computer the singular value decomposition of $\mathbf{A}$
- Find elements of $\mathbf{F}$ as the coefficients of the column in $\mathbf{V}$ corresponding to the smaller Eigenvalue
- Enforce rank constraint $r = 2$
- Un-normalize $\mathbf{F}$

In Defence of the 8-point Algorithm

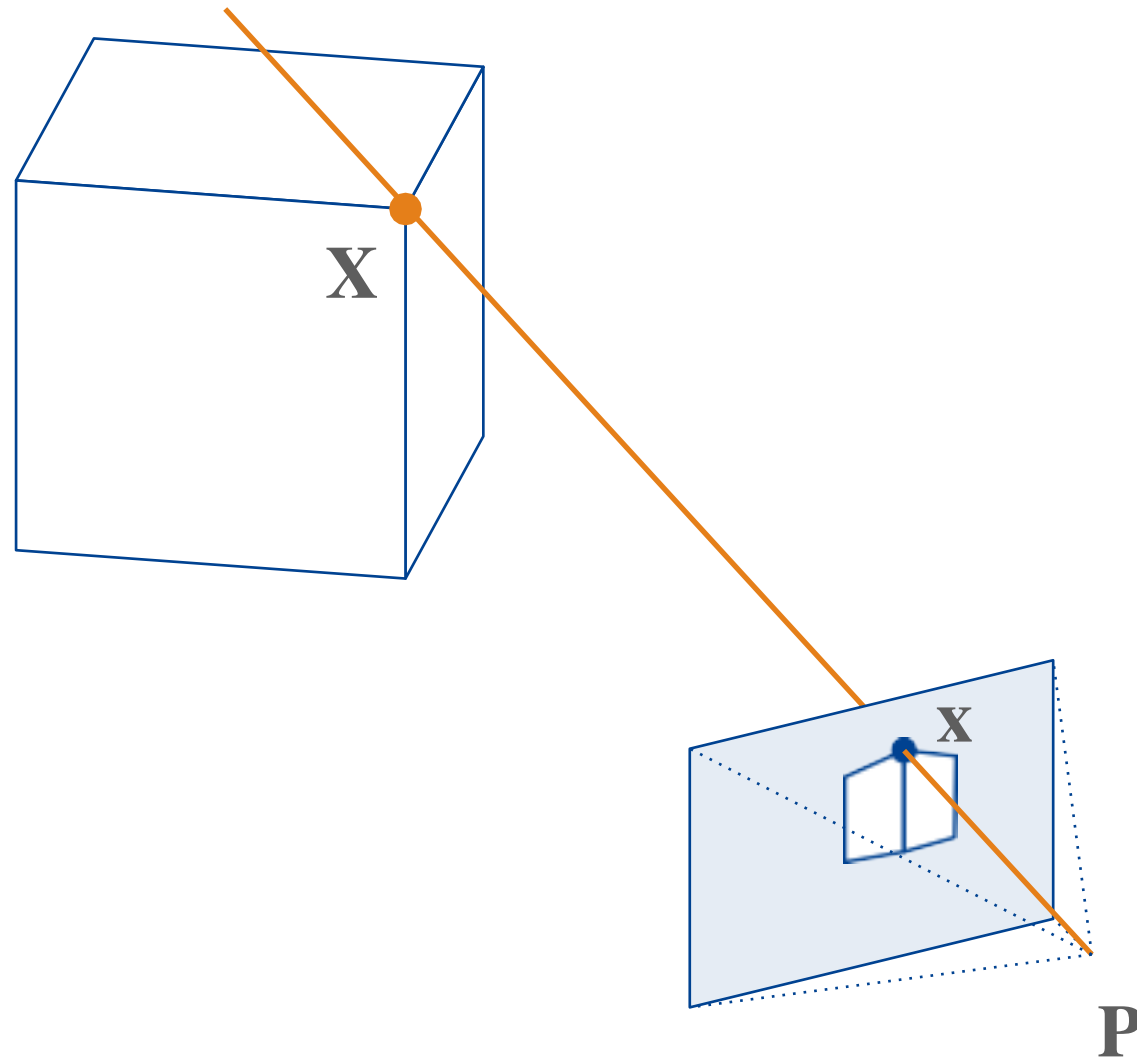
Richard I. Hartley,
GE-Corporate Research and Development,
Schenectady, NY, 12309,
email : hartley@crd.ge.com

What can we do with this now?

- Compute sparse 3D point clouds from images and point feature correspondences
- Use the essential or fundamental matrix to search more point correspondences under the strong epipolar constraint
 - Given a point in the first image, we only need to search along a line in the second image
- Actually, we'd like to match each pixel in one image to the other image to compute a dense 3D point cloud of the object surface („**Dense stereo reconstruction**“)

Triangulation

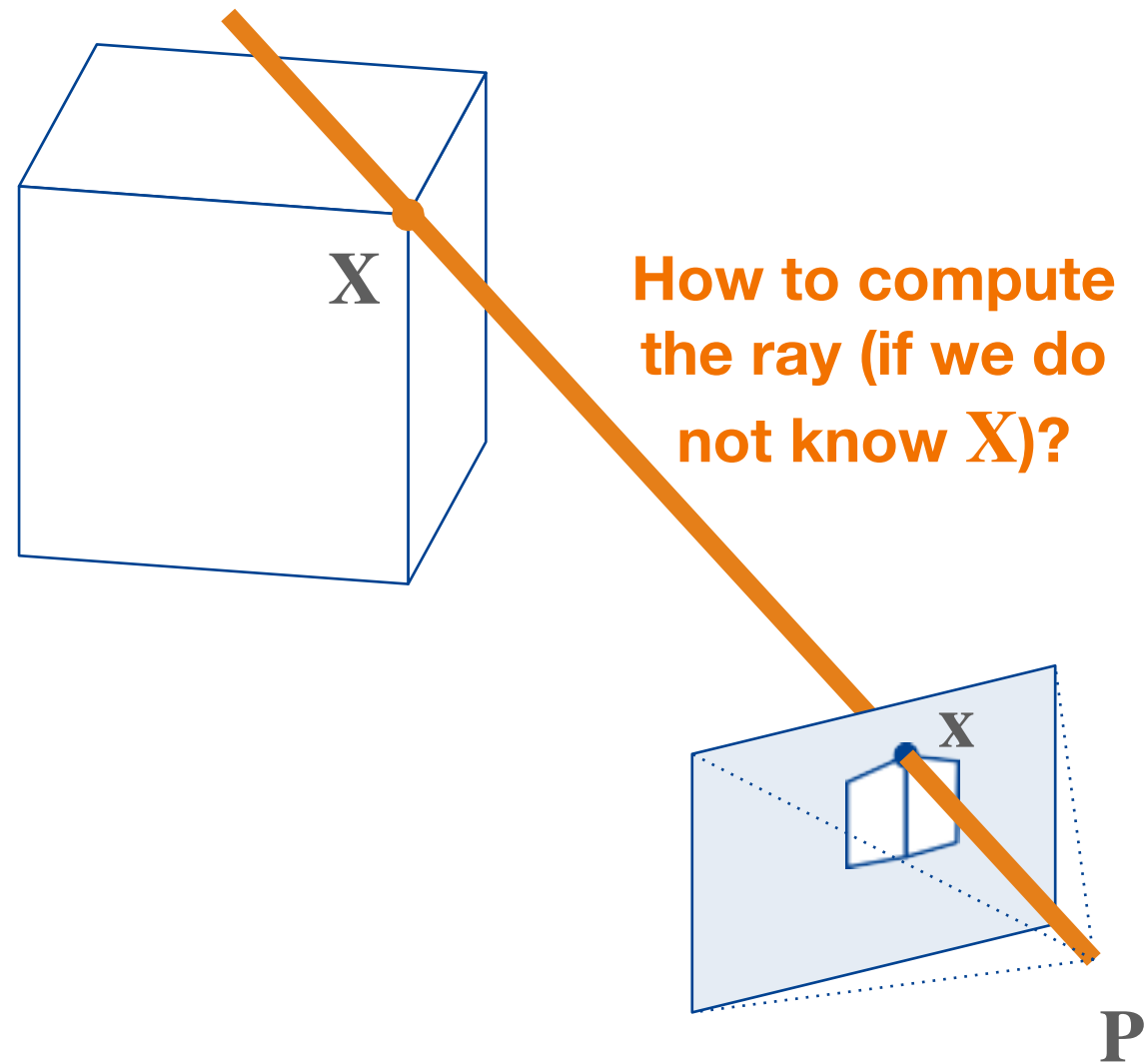
Finding the 3D position of an observed point



- Camera with projection matrix $\mathbf{P}$ observes 3D point $\mathbf{X}$ as

$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

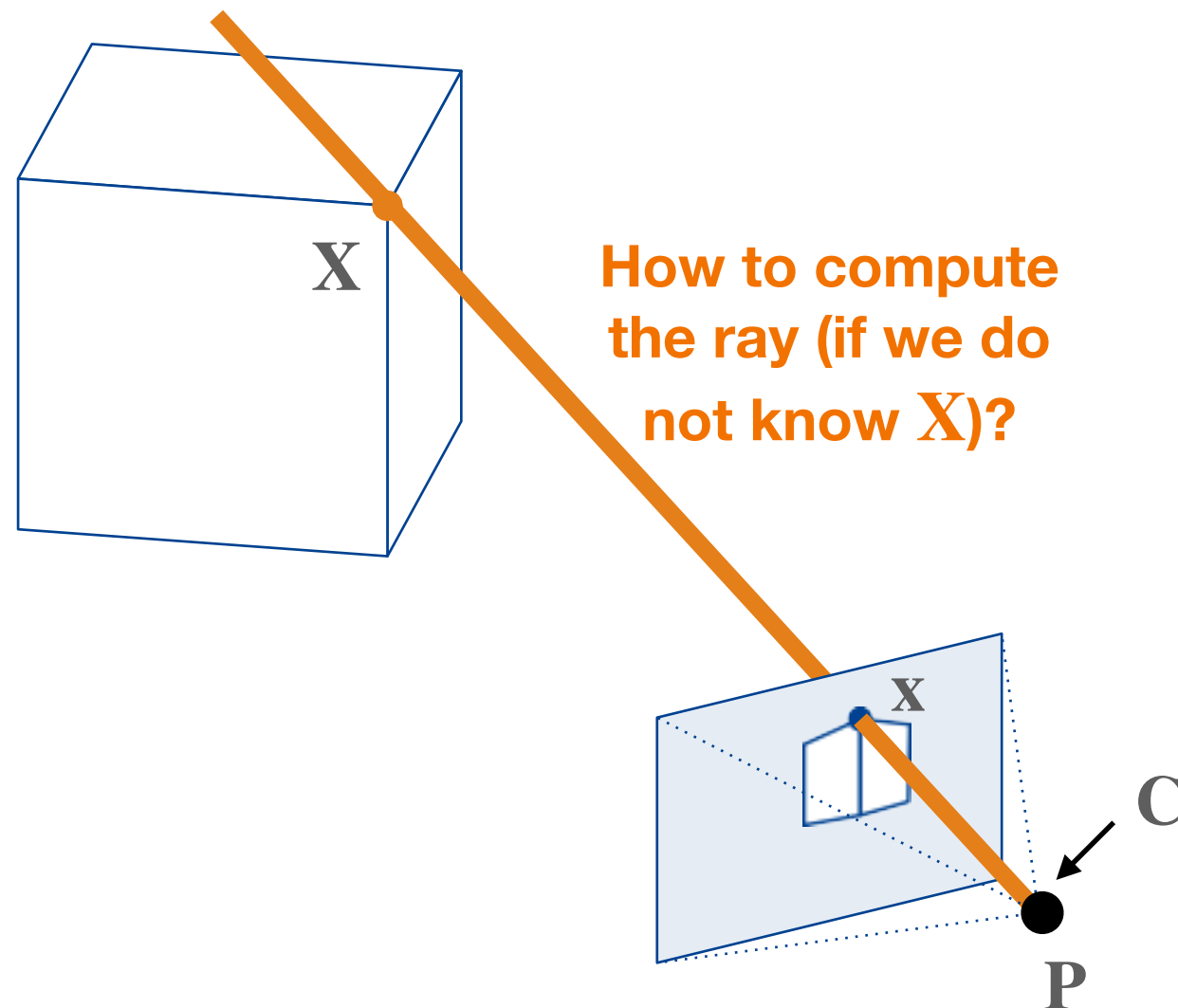
Finding the 3D position of an observed point



- Camera with projection matrix **P** observes 3D point **X** as

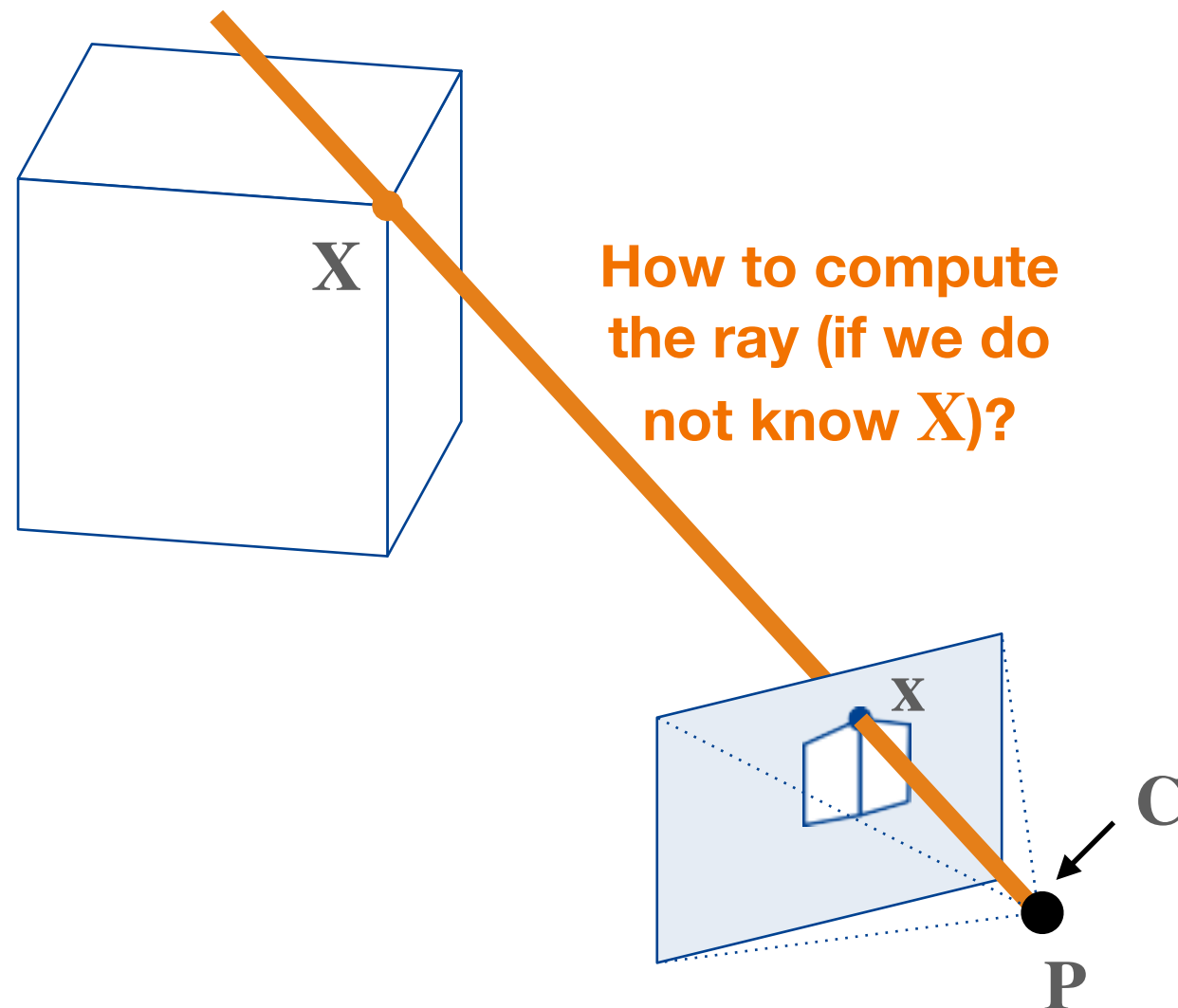
$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

Finding the 3D position of an observed point



- Ray passes through projection center C
- Ray passes through x back projected to the world coordinate system
- Could be done using the pseudoinverse:
$$? = \mathbf{P}^+ \mathbf{x}$$
- But doesn't this give us X already?

Finding the 3D position of an observed point



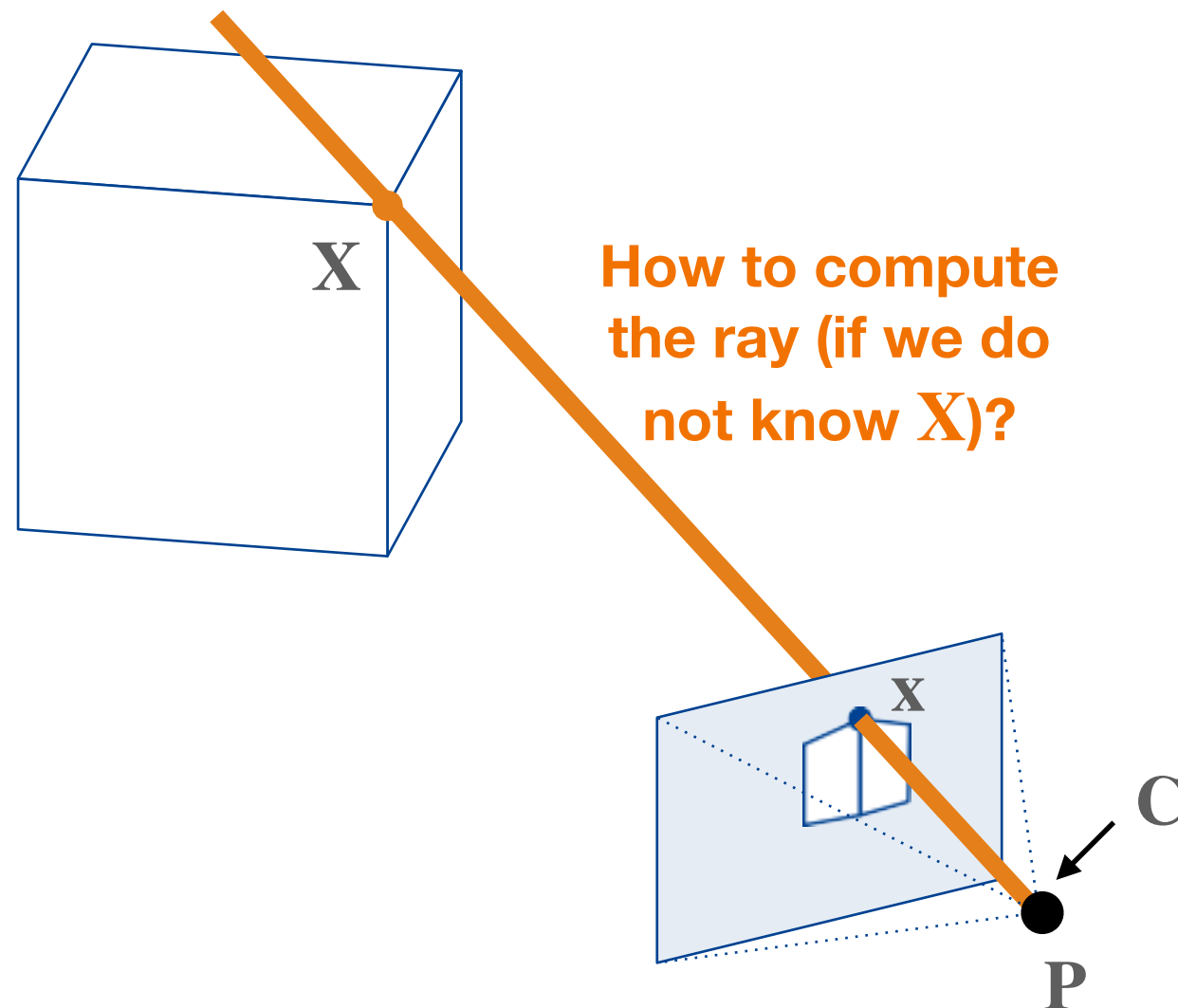
Given the 3D point, we obtain the homogenous coordinate

$$\mathbf{x} = \mathbf{P}\mathbf{X} \simeq \lambda \mathbf{P}\mathbf{X}$$

which expands to

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Finding the 3D position of an observed point

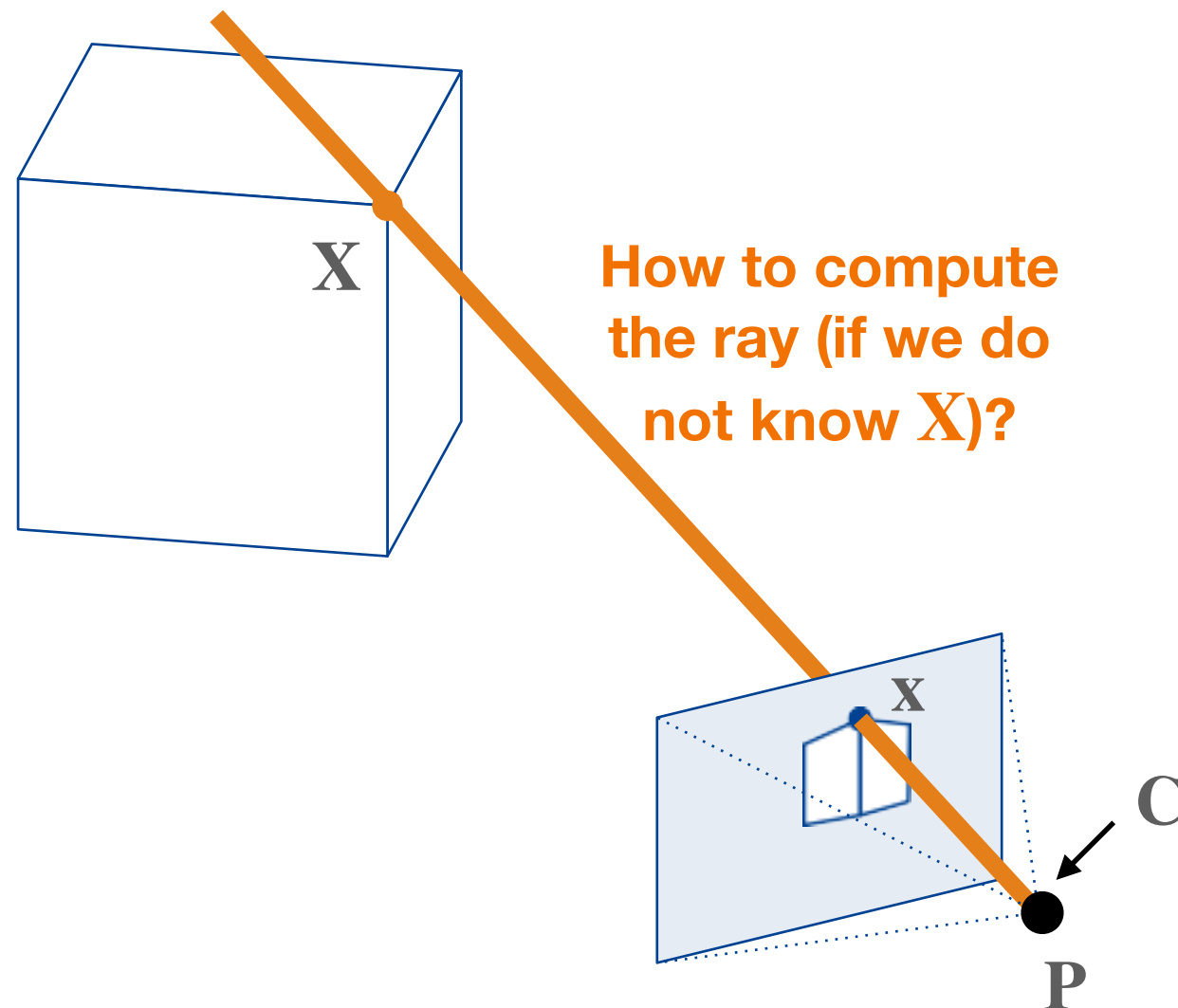


We want to estimate the unknowns X, Y, Z and arrive at a formulation

$$\mathbf{A}\mathbf{X} = 0$$

But how to deal with λ ?

Finding the 3D position of an observed point



We want to estimate the unknowns X, Y, Z and arrive at a formulation

$$\mathbf{A}\mathbf{X} = 0$$

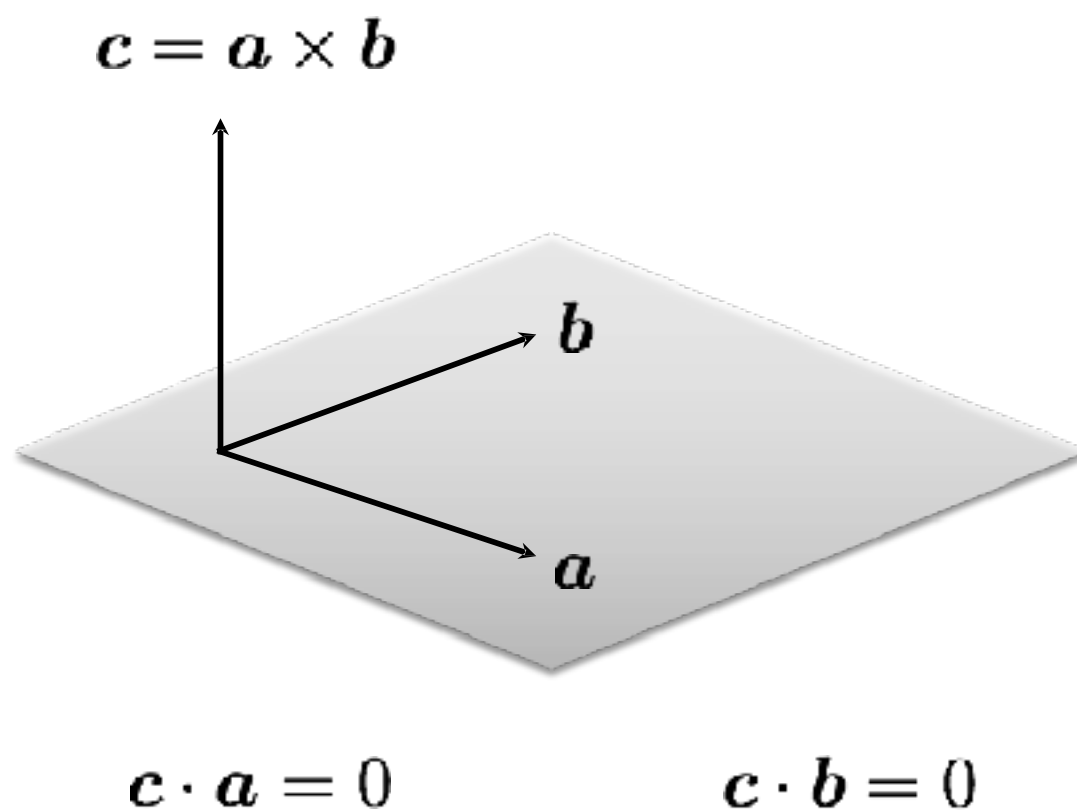
But how to deal with λ ?

Solution: Check for collinearity of $\mathbf{x}$ and $\mathbf{P}\mathbf{X}$ using the cross product

$$\mathbf{x} \times \mathbf{P}\mathbf{X} = 0$$

Cross product

Given two vectors, returns a vector perpendicular to both



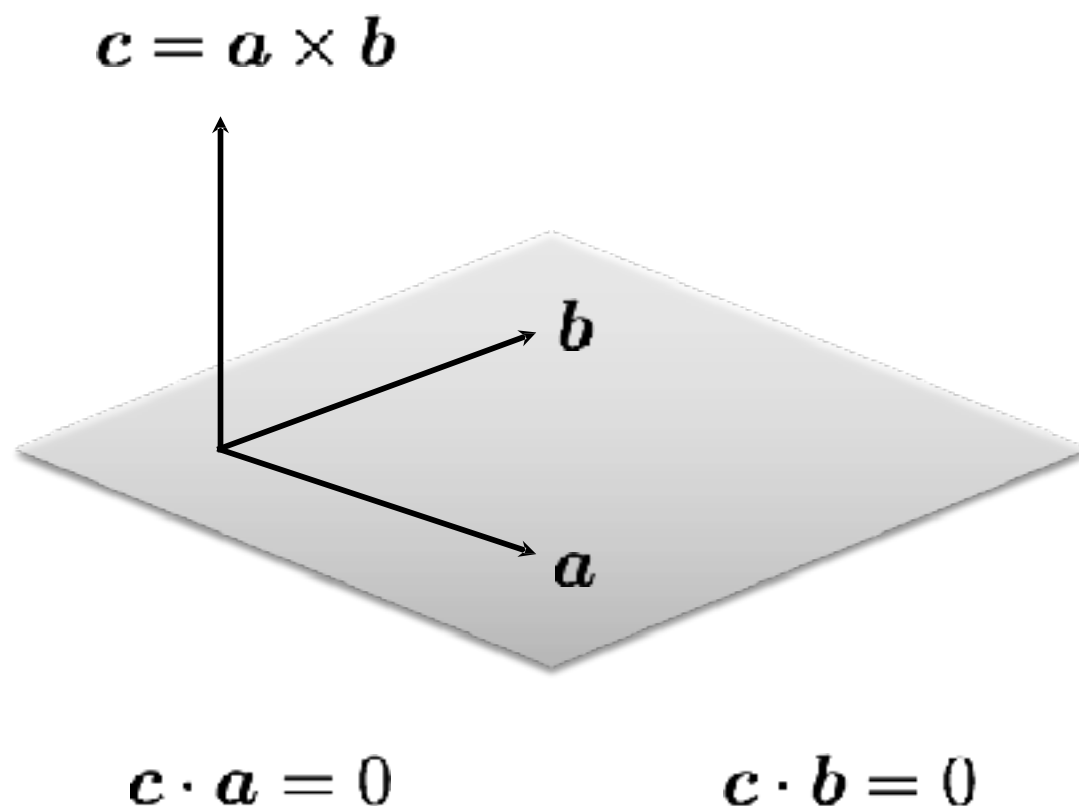
$$a \times b = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

$$a \times a = 0$$

Cross product

Can be written as a matrix multiplication using the skew symmetric matrix

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$



Building the linear system

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} \text{---} & \mathbf{p}_1^\top & \text{---} \\ \text{---} & \mathbf{p}_2^\top & \text{---} \\ \text{---} & \mathbf{p}_3^\top & \text{---} \end{bmatrix} \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} \mathbf{p}_1^\top \mathbf{X} \\ \mathbf{p}_2^\top \mathbf{X} \\ \mathbf{p}_3^\top \mathbf{X} \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \times \begin{bmatrix} \mathbf{p}_1^\top \mathbf{X} \\ \mathbf{p}_2^\top \mathbf{X} \\ \mathbf{p}_3^\top \mathbf{X} \end{bmatrix} = \begin{bmatrix} y\mathbf{p}_3^\top \mathbf{X} - \mathbf{p}_2^\top \mathbf{X} \\ \mathbf{p}_1^\top \mathbf{X} - x\mathbf{p}_3^\top \mathbf{X} \\ x\mathbf{p}_2^\top \mathbf{X} - y\mathbf{p}_1^\top \mathbf{X} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Given one projected point pair $(\mathbf{x}, \mathbf{X})$, how many constraints (equations) do we get?

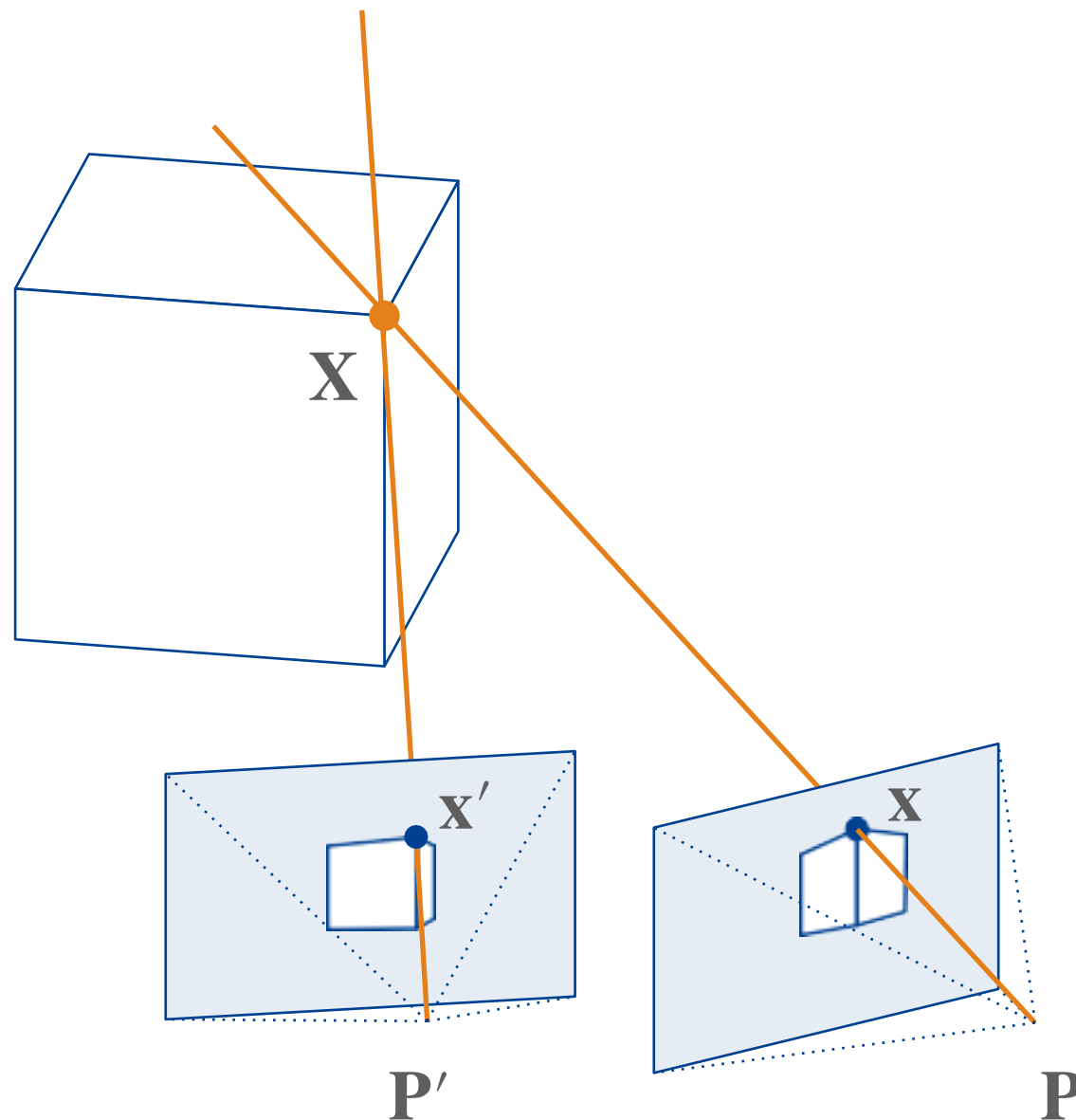
Building the linear system

Since row 3 in the cross product is a linear combination of rows 1 and 2, we only get 2 constraints from the observation of X in one camera.

$$\begin{bmatrix} y\mathbf{p}_3^\top - \mathbf{p}_2^\top \\ \mathbf{p}_1^\top - x\mathbf{p}_3^\top \end{bmatrix} \mathbf{X} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{A}_i \mathbf{X} = \mathbf{0}$$

Finding the 3D position of an observed point



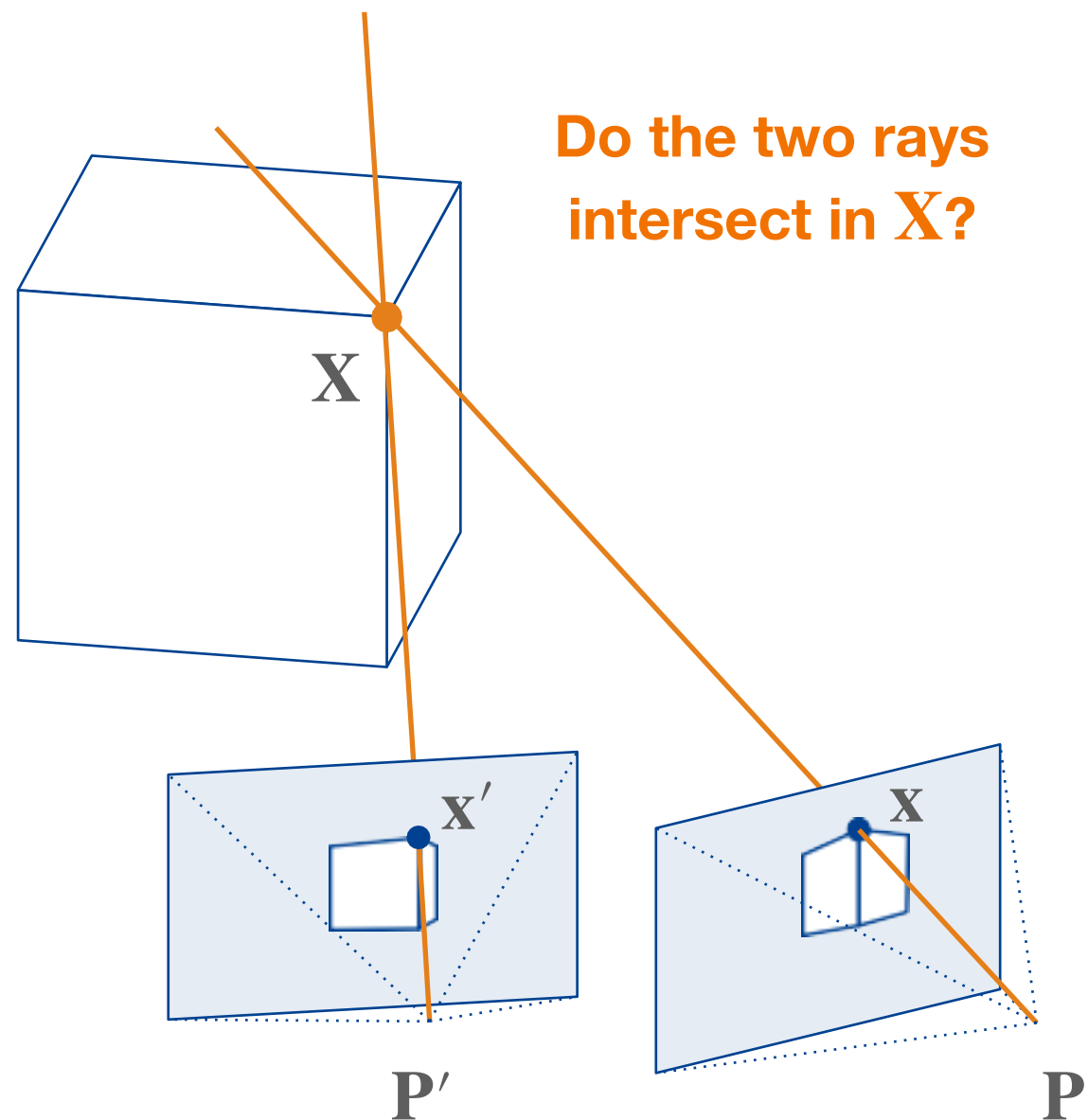
- Camera with projection matrix $\mathbf{P}$ observes 3D point $\mathbf{X}$ as

$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

- Second camera with projection matrix $\mathbf{P}'$ observes the same point as

$$\mathbf{x}' = \mathbf{P}'\mathbf{X}$$

Finding the 3D position of an observed point



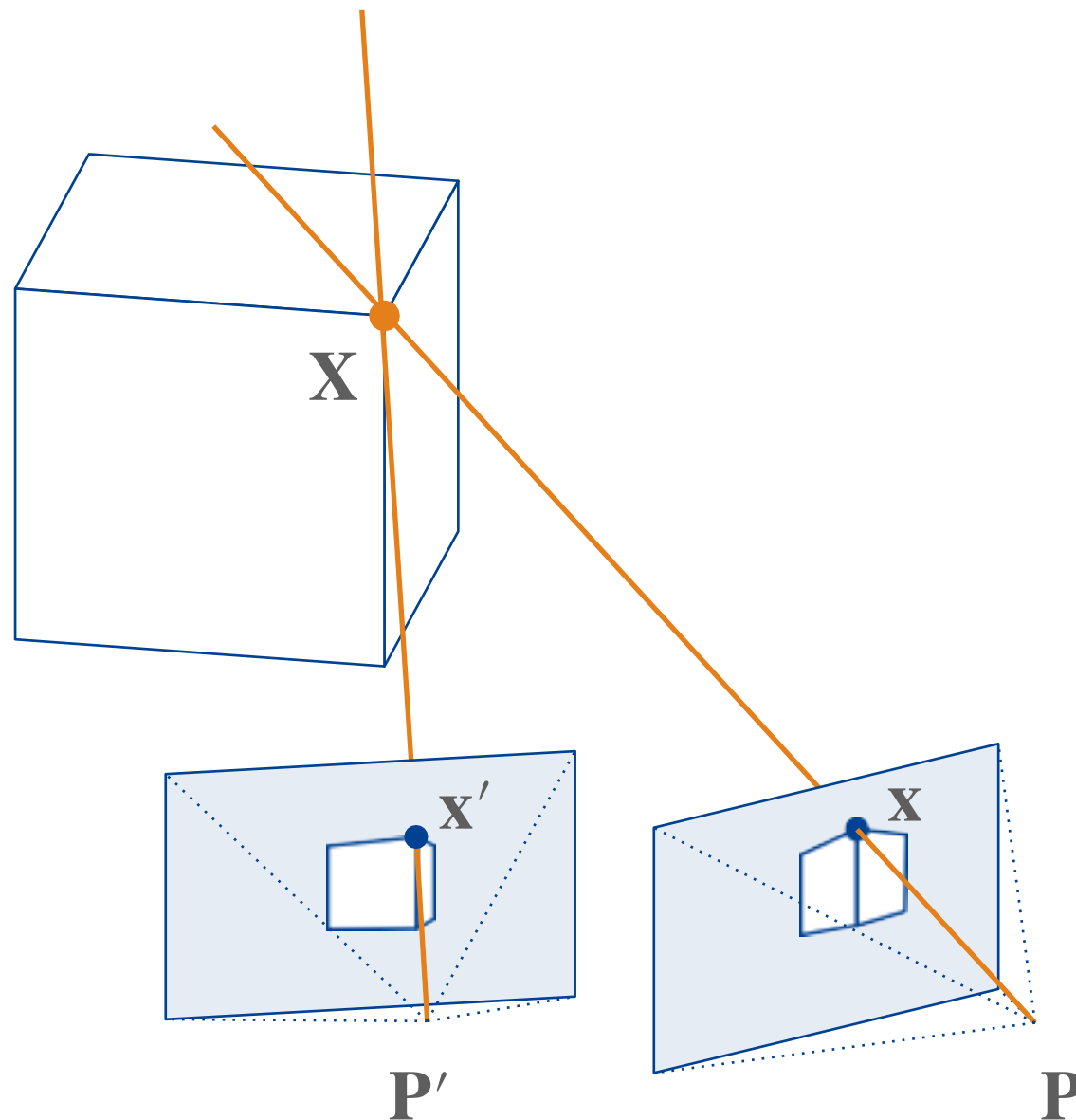
- First camera with projection matrix $\mathbf{P}$ observes 3D point $\mathbf{X}$ as

$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

- Second camera with projection matrix $\mathbf{P}'$ observes the same point as

$$\mathbf{x}' = \mathbf{P}'\mathbf{X}$$

Finding the 3D position of an observed point



$$\mathbf{A}\mathbf{X} = \mathbf{0}$$

$$\begin{bmatrix} y\mathbf{p}_3^\top - \mathbf{p}_2^\top \\ \mathbf{p}_1^\top - x\mathbf{p}_3^\top \\ y'\mathbf{p}'_3^\top - \mathbf{p}'_2^\top \\ \mathbf{p}'_1^\top - x'\mathbf{p}'_3^\top \end{bmatrix} \mathbf{X} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Find a least squares solution using singular value decomposition.